

SPO:

$$m \frac{d^2 x}{dt^2} + kx = 0 \Rightarrow \frac{d^2 x}{dt^2} + \left(\frac{k}{m}\right)x = 0$$

$$F = -kx = -\frac{dp}{dt} = m \frac{dv}{dt} = m \frac{d^2 x}{dt^2}$$

$$\therefore m \frac{d^2 x}{dt^2} + kx = 0$$

• Has a classical solution

$$x(t) = c_1 e^{i\omega t} + c_2 e^{-i\omega t}$$

• Classical Energy Ham $H = T + V = \frac{1}{2} \frac{p^2}{m} + \frac{1}{2} m\omega^2 x^2$

$$-\nabla V = F \Rightarrow \frac{d}{dx} V = kx \Rightarrow \frac{1}{2} kx^2$$

$$\omega^2 = \frac{k}{m} \Rightarrow k = m\omega^2 \Rightarrow V = \frac{1}{2} m\omega^2 x^2$$

• Quantized (Dirac): $[\hat{x}, \hat{p}] = i$

$$\hat{a} = \sqrt{\frac{m\omega}{2}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right), \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2}} \left(\hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

$$\hookrightarrow [\hat{a}, \hat{a}^\dagger] = 1$$

$$\hat{a} + \hat{a}^\dagger = \sqrt{2m\omega} \hat{x} \Rightarrow \hat{x} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2m\omega}}$$

• replace m to $\hbar$ to obtain

$$H = \omega \left(a a^\dagger + \frac{1}{2} \right) / \quad \hat{N} = a^\dagger a$$

$$\begin{cases} N|n\rangle = n|n\rangle \\ a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \\ a|n\rangle = \sqrt{n}|n-1\rangle \end{cases}$$

$$|1\rangle = a^\dagger|0\rangle$$

$$|2\rangle = a^\dagger a^\dagger|0\rangle$$

$\vdots$

$$\therefore \hat{N}|n\rangle = \underbrace{a^\dagger a}_{1} (a_n^\dagger \dots a_1^\dagger) |0\rangle$$

$$\begin{aligned} &= a^\dagger (a a + 1) a_{n-1}^\dagger \dots a_1^\dagger |0\rangle \\ &= a^\dagger (a_n^\dagger a a_{n-1}^\dagger \dots) |0\rangle + a^\dagger a_{n-1}^\dagger \dots a_1^\dagger |0\rangle \\ &= a^\dagger (a_n^\dagger a a_{n-1}^\dagger \dots a_1^\dagger |0\rangle + |n\rangle) \end{aligned}$$

$n-1$ more times

$$= a^\dagger a_n^\dagger \dots a_1^\dagger a |0\rangle + n|n\rangle$$

$$a|0\rangle = 0$$

$$\therefore \hat{N}|n\rangle = n|n\rangle$$

• Heisenberg picture: $i \frac{d}{dt} a(t) = [a(t), H(t)]$
 $= \omega a$

$$\therefore a(t) = a(0) e^{-i\omega t}$$

$$\square \phi = 0$$

$$(-\partial_t^2 + \vec{\nabla}^2) \phi = 0$$

Has plane wave solutions

$$\phi(x) = a_p(t) e^{i \vec{p} \cdot \vec{x}} \Rightarrow (\partial_t^2 + \vec{p} \cdot \vec{p}) a_p(t) = 0$$

$$\phi(x, t) = \int \frac{d^3 p}{(2\pi)^3} [a_p(t) e^{i \vec{p} \cdot \vec{x}} + a_p^*(t) e^{-i \vec{p} \cdot \vec{x}}]$$

$$|\vec{p}|^2$$

$$a_p(t) = a_p(0) e^{-i |\vec{p}| t}$$

$$\therefore \phi(x) = \int \frac{d^3 p}{(2\pi)^3} (a_p e^{i \vec{p} \cdot \vec{x}} + a_p^* e^{-i \vec{p} \cdot \vec{x}})$$

* Classically H above is the Dirac H

$$H = \int \frac{d^3 p}{(2\pi)^3} \quad \text{would be } \int d^3 x \text{ but we are in Fourier space.}$$

$$H_0 = \int \frac{d^3 p}{(2\pi)^3} \omega_p \left(\vec{a}_p \cdot \vec{a}_p + \frac{1}{2} \right)$$

$$[a_k, a_p] = (2\pi)^3 \delta^3(k - p)$$

$$a_p^\dagger = \frac{1}{\sqrt{2\omega_p}} |p\rangle \quad |p\rangle = \sqrt{2\omega_p} a_p^\dagger |0\rangle$$

$$\langle 0 | 0 \rangle = 1$$

$$\begin{aligned} \langle \vec{p} | \vec{k} \rangle &= 2 \sqrt{\omega_p \omega_k} \langle 0 | a_p a_k^\dagger | 0 \rangle \\ &= 2 \sqrt{\omega_p \omega_k} (2\pi)^3 \delta^3(\vec{p} - \vec{k}) \\ &= 2 \omega_p (2\pi)^3 \delta^3(\vec{p} - \vec{k}) \end{aligned}$$

$$\begin{aligned} \mathbb{1} |\vec{p}\rangle &= |\vec{p}\rangle \\ \langle \vec{k} | \mathbb{1} |\vec{p}\rangle &= \langle \vec{k} | \vec{p} \rangle \end{aligned}$$

$$\int \frac{d^3 p}{(2\pi)^3} |\langle \vec{k} | \mathbb{1} |\vec{p}\rangle| = 2 \omega_k$$

$$\int \frac{d^3 p}{(2\pi)^3} \left(\frac{1}{2\omega_k} \right) \langle \vec{k} | \mathbb{1} |\vec{p}\rangle = 1$$

Since $|\psi\rangle = \sum_n \psi_n |n\rangle$ in QM Here we have

$$|\psi\rangle = \int d^3x \psi(x) |x\rangle$$

$$\therefore \langle y|\psi\rangle = \int d^3x \psi(x) \langle y|x\rangle = \psi(y)$$

$$\therefore \langle y|x\rangle = \delta^3(y-x)$$

$$\psi(p) = \int d^3x e^{iPx} \psi(x)$$

$$\langle p|\psi\rangle = \int d^3x e^{iPx} \langle x|\psi\rangle$$

$$\langle p| = \int d^3x e^{iPx} \langle x|$$

$$\langle \vec{p}|\vec{y}\rangle = \int d^3x e^{iPx} \langle x|y\rangle$$

$$\langle p|y\rangle = e^{iPy}$$

$$\langle 0|\sqrt{2\omega_p} a = \sqrt{2\omega_p} \langle p|$$

$$\begin{aligned} \langle x|\psi\rangle &= \int \frac{d^3p}{(2\pi)^3} e^{-iPx} (\langle 0|a_p)|\psi\rangle \\ &= \int \frac{d^3p}{(2\pi)^3} \langle x|a_p b\rangle b| \psi\rangle \end{aligned}$$

Normalize: $(\langle 0|a_p)|y\rangle = e^{iPy}$ and

$$" \psi(p) " = \langle 0|a_p|\psi\rangle$$

multiply by $\frac{2\omega_p}{2\omega_p}$

$$\langle x|4\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{\langle x|p\rangle \langle p|4\rangle}{2\omega_p}$$

$$\therefore \langle x|4\rangle = \underbrace{\int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle \langle p|}_{1} (\langle x|4\rangle)$$

(free-field)

$$\phi_0(\vec{x}) = \int d^3p (a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}})$$

$$\langle \vec{p} | \hat{\phi}_0(\vec{x}) | 0 \rangle$$

* $\hat{\phi}_0(x)$ acts on the vacuum $|0\rangle$ and we project out a momentum component $\langle \vec{p} |$.

$$\begin{aligned} \langle \vec{p} | \hat{\phi}_0(\vec{x}) | 0 \rangle &= \sqrt{2\omega_p} \langle 0 | a_p \left(\int d^3k (a_k e^{i\vec{k}\cdot\vec{x}} + a_k^\dagger e^{-i\vec{k}\cdot\vec{x}}) \right) | 0 \rangle \\ &= \sqrt{2\omega_p} \int \frac{d^3k}{(2\pi)^3} \sqrt{2\omega_k} (2\pi)^3 \delta(\vec{k} - \vec{p}) e^{-i\vec{p}\cdot\vec{x}} \\ &= e^{i\vec{p}\cdot\vec{x}} \end{aligned}$$

$$\langle \vec{p} | \hat{\phi}_0(\vec{x}) | 0 \rangle = \langle \vec{p} | \vec{x} \rangle$$

$\therefore$

$$\boxed{\hat{\phi}_0(\vec{x}) | 0 \rangle = |\vec{x}\rangle}$$

* The free field $\hat{\phi}_0(\vec{x})$ creates a particle of $|\vec{x}\rangle$

$$[\mathcal{H}_0, \phi_0(\underline{x})] = -i\partial_t \phi_0(\underline{x})$$

$$* \psi(x) = \langle x|4\rangle = \langle 0 | \phi_0(x) | 4 \rangle$$

$$i\partial_t \psi(x) = i\partial_t (\langle 0 | \phi_0(x) | 4 \rangle)$$

$$(\partial_t^2 - \vec{\nabla}^2 + m^2)\phi_0 = 0$$

$$i\partial_t \psi(x) = \langle 0 | i\partial_t \phi_0(x) | 4 \rangle$$

$$= \langle 0 | \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} i\omega_k \left(a_k e^{i\omega_k t - i\vec{k}\cdot\vec{x}} + -i\omega_k a_k^\dagger e^{-i\omega_k t + i\vec{k}\cdot\vec{x}} \right) | 4 \rangle$$

$$= \langle 0 | \int \frac{d^3k}{(2\pi)^3} \frac{\omega_k}{\sqrt{2\omega_k}} \left(a_k e^{i\vec{k}\cdot\vec{x}} - a_k^\dagger e^{-i\vec{k}\cdot\vec{x}} \right) | 4 \rangle$$

$$\omega_k = \sqrt{\vec{k}^2 + m^2}$$

$$= \langle 0 | \sqrt{m^2 - \vec{\nabla}^2} \phi_0(x) | 4 \rangle$$

$$= \sqrt{m^2 - \vec{\nabla}^2} \langle 0 | \phi_0(x) | 4 \rangle = \sqrt{m^2 - \vec{\nabla}^2} \psi(x)$$

$$i\partial_t \psi(x) = \left(m - \frac{\vec{\nabla}^2}{2m} + \mathcal{O}\left(\frac{1}{m^2}\right) \right) \psi(x)$$

(low energy / High mass limit)

* Remove the rest mass mc^2

$$-i\partial_t \psi(x) = -\frac{\vec{\nabla}^2}{2m} \psi(x)$$

$$[\phi(x), \phi(y)] = 0$$